João Victor Mendes

Production Support Engineer (PSE) / SRE | Cloud · Observability · Database Reliability
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PROFESSIONAL SUMMARY

Production Support Engineer (PSE) / SRE with ∼4 years of experience sustaining corporate fleets of 500+ environments (1,500+ users) and distributed cloud platforms (FinTech and AI). Strong focus on relational database internals (PostgreSQL, Aurora MySQL), saturation diagnostics, P1/P2 incident response, and detection engineering with MTTD reduction validated in incident replay. Author and operator of an active public hydrological platform in production (>3.2M records).

TECHNICAL SKILLS

- **Cloud & Infrastructure:** AWS (Aurora MySQL/RDS, EC2, EKS, SSM, CloudFront, Route 53, ACM), Kubernetes (Karpenter, ArgoCD GitOps, Pod Topology Spread Constraints), Docker, Linux (systemd, boot/network troubleshooting), Nginx.
- **Observability & SRE:** New Relic (NRQL, NerdGraph API, baseline/alerting), Incident Response (P1/P2), Post-Mortem / RCA (ITIL), Detection Engineering & Runbooks.
- **Databases:** AWS Aurora MySQL (Performance Insights, connection pooling), PostgreSQL (WAL, replication slots, unlogged tables, pg_class/pg_toast catalogs, EXPLAIN ANALYZE), Firebird.
- **Backend & Orchestration:** Python, Airflow, Bash, PowerShell.
- **IaC, Automation & Testing:** Terraform, SQL, PL/pgSQL, Playwright, Git.
- **Security & Access:** pg_hba access hardening (SCRAM-SHA-256 vs. MD5, scoped IP ranges), audit log analysis, secret management (SSM Parameter Store SecureString).
- **Languages:** English (B2 Certified by BRASAS; daily technical documentation and incident response), Portuguese (Native).

PROFESSIONAL EXPERIENCE

Monest | FinTech & AI Platform — Aug 2026 – Present
Production Support Engineer (PSE) / SRE — Remote, Brazil

Sustaining reliability and observability for a cloud platform powering microservices, VoIP dialers, and AI agents across backend, infrastructure, and GitOps.

- **Database Saturation Diagnostics & Capacity (Aurora MySQL):** Diagnosed critical CPU and connection saturation on a production Aurora cluster during an asynchronous queue backlog, isolating root cause to unbatched writes (majority of writer load in telemetry) and microservice connection pools exceeding cluster concurrency limits; defined per-service pool limits and recommended RDS Proxy adoption.
- **Detection Engineering & Alerting Calibration (New Relic / Terraform / Metabase):** Audited hundreds of alert conditions via GraphQL, isolating an unsegmented policy and mathematically flawed baselines where healthy traffic dilution delayed critical detections by hours. In Metabase backtesting, distinct-session counting surfaced incidents hours earlier and slashed alert noise by ∼60%, calibrating compound rules in Terraform to reduce noise without coverage loss.
- **External Integration Root Cause (API / Observability):** Diagnosed intermittent failures in an external provider integration, tracing root cause to schema validation within a `try` block masking HTTP 200 responses as 500 errors; correlated HTTP telemetry against error logs to isolate false positives, resulting in a >99% drop in integration errors following upstream schema fix.
- **Data Pipeline Incident Response (Airflow / Docker / SSM):** Led post-mortem for pipeline downtime caused by disk exhaustion, restoring host access via AWS Systems Manager (SSM) under least-privilege IAM roles; implemented an NRQL deadman alert (via Terraform with runbook) that, in incident replay, demonstrated an MTTD reduction from ∼2h to ∼10 min, and resolved an IaC disk rule where partition averaging masked root disk saturation.
- **Scheduler Reliability & Host Hardening (Systemd / Linux):** Diagnosed root cause of silent scheduler service interruptions caused by health checks asserting reverse-proxy HTTP 200s rather than job execution; during controlled reboot, resolved boot races via `After=network-online.target` preventing cascading failures, extended resilient systemd restart policies across core host services, and configured Slack failure alerts via `OnFailure` with SSM credentials.
- **High Availability & K8s Resiliency (EKS / ArgoCD):** Mitigated compute node disruptions across the asynchronous workload stack following Spot node terminations; restructured GitOps manifests via ArgoCD enforcing strict node-level `topologySpreadConstraints` (`DoNotSchedule`) to prevent single-node bin-packing, enabling cross-node high availability in production across Spot compute capacity.

Pedroso Automação | B2B fiscal compliance ERP SaaS Jun 2024 – Aug 2026
Technical Support Engineer (L2/L3) União da Vitória, PR

Direct L2/L3 support across 500+ corporate environments (1,500+ end-users), maintaining an average of 80+ technical tickets/week on solo rotations and resolving P1/P2 incidents.

- **WAL Retention Investigation & Replication Guardrails:** Traced recurring fleet-wide disk exhaustion to orphaned slots in `pg_replication_slots` retaining WAL for deactivated logical replication subscribers; built PowerShell automation dropping inactive slots (>2GB or >30 min without consumer) and enforced `max_slot_wal_keep_size` on PG13+, with zero recurrence of disk exhaustion across environments where the routine was deployed.
- **Fleet Standardization & Governance (PostgreSQL):** Automated PostgreSQL instance provisioning via standardized scripts (cutting setup from 2h to 7 min, ~3 servers/week) and applied authentication hardening: SCRAM-SHA-256 by default and MD5 restricted by IP in `pg_hba.conf` for legacy clients.
- **Cluster Recovery & Crash-Recovery Orchestration (PostgreSQL):** Reduced host recovery time across 30 databases from ~3h to <5 min via wave-based parallel startup with concurrency bounds, safe orphaned `postmaster.pid` cleanup, and pre-flight WAL validation, and gating `pg_resetwal` to require operator judgment due to data-loss risks.
- **Data Growth Diagnostics & Restore Optimization:** Diagnosed slow restore times by auditing PostgreSQL catalogs and TOAST storage; traced anomalous ~1GB/month growth to serialized paths in audit logs (establishing a purge routine) and cut restore time from 17 to 7 min (`pg_restore -j 4`).
- **70M-Record ETL Pipeline Engineering:** Designed autonomous pipeline processing a 15GB+ government dataset (70M+ records) in PL/pgSQL and PowerShell with `unlogged` tables, cutting processing time by 60% (~4h to ~1h30) and surfacing ~200 qualified leads that shaped Q3 B2B expansion.
- **Critical Data Recovery & Reverse Engineering (Firebird):** Protected critical corporate inventory (10,000+ items) by reverse-engineering an undocumented legacy database using AI, reprocessing 20 years of logs with zero data loss verified against accounting and fiscal balances, delivered on the day of first schema access.
- **Runbooks & LLM-Assisted Triage:** Authored 20+ ITIL runbooks and training materials enabling L1 engineers to autonomously resolve daily escalations; implemented LLM-assisted triage injecting database schema and fiscal rules, reducing diagnosis time from 30+ to <5 min.

Girafa Comunicação Interativa | Digital Agency Sep 2022 – Apr 2024
AWS Support Engineer | Cloud Infrastructure & Web Applications Remote, Brazil

Sustained AWS cloud infrastructure and technical support for 70+ corporate clients in a multidisciplinary team.

- **AWS Cloud Operations, DNS & Release Governance:** Administered cloud infrastructure and corporate DNS on AWS (EC2, RDS MySQL 8, CloudFront, ACM) and Route 53 with external registrars (Registro.br, Locaweb); executed 15+ DNS zone migrations with prior TTL reduction and zero reported email or web downtime, isolated failure domains between CDN, DNS, and email authentication (SPF, DKIM, DMARC), and coordinated staged Git deployments with freeze and pre-production validation.
- **Security Incident Response & Technical Liaison:** Investigated security breach auditing Git commits, CMS roles, and access logs to isolate the attack vector, containing the threat within 24h of detection via session revocation and credential rotation across 10+ environments; served as technical liaison in English with US and EU vendors during escalations and root-cause analysis.

PRODUCTION ENGINEERING PROJECTS

Monitoramento do Rio Iguaçu | Hydrological Platform & Flood Simulation Jun 2026 – Present
Author and operator (in production since Sep 2026) rioiguacu.jornalocomercio.com

- **Multi-Source Telemetry Ingestion & Data Reliability:** Autonomous pipeline (15-min systemd timer) integrating 5 basin telemetric stations and the ANA historical series (since 1930), alongside Copel and ONS operational data, accumulating >3.28 million immutable records; implemented strict schema validation and data freshness checks that isolate upstream layout or payload breaks without corrupting the historical time series.
- **Operational Reliability Under Peak Traffic:** Served 2.7M+ requests on the Jornal O Comércio news domain since deployment (Sep 2026) and absorbed a peak of 43k req/hour during a flood event (5.5m alert level, >20x baseline traffic) without recorded downtime. Handled peak traffic on a 4 GB RAM VPS (2 vCPUs) via pre-computed GeoJSON layers served by Nginx (ETag/Cache-Control) and singleflight concurrency in the Python backend against DEM compute spikes; operation features 120+ immutable releases, symlink rollbacks, E2E testing with Playwright, and systemd isolation (`ProtectSystem=strict`).

EDUCATION & AWARDS

FGV | Data Science & Economics (2020–2022, 4 semesters completed) Rio de Janeiro, Brazil
Full-ride national merit scholarship (OBMEP). Focus on probability, applied statistics, and programming.

IFPR | Systems Analysis & Development (2023–2024, 3 semesters completed) União da Vitória, Brazil
Focus on algorithms, database modeling, and computer networking.

Silver Medalist | Brazilian Public School Mathematics Olympiad (OBMEP, 2016) National
Nationally ranked in the top 0.01% among 18M+ students; Bronze Medalist (2014) and 2x Honorable Mentions.